

Sai Pottipothula

Minneapolis, MN | +1 (201) 275-9949 | pbsn99@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer experienced in building scalable backend systems using Java and C# (.NET 6+) across banking and financial domains. Skilled in developing REST APIs, integrating distributed systems, and improving application performance and reliability. Strong in debugging, testing, and contributing across the full SDLC lifecycle, delivering maintainable and high-quality enterprise solutions

TECHNICAL SKILLS

Programming Languages: Java, C#, Python, JavaScript (ES6+), SQL
Backend: .Net 6+, ASP.NET Core, Spring Boot, Spring Security (JWT, OAuth2), JPA/Hibernate, Entity Framework Core, REST APIs, API Integration
Frontend: React.js, HTML, CSS, Tailwind CSS, Redux Toolkit, TypeScript
Database: SQL Server, PostgreSQL, MySQL, SQL Query Optimization, Indexing
Cloud & DevOps: AWS (EC2, S3, Lambda, API Gateway, IAM), Docker, CI/CD Pipelines (GitHub Actions, Jenkins)
Testing & Observability: JUnit, xUnit, Integration Testing, Postman, Swagger, SonarQube, Prometheus, Grafana, ELK Stack
Tools & Practices: Git, Jira, Confluence, IntelliJ IDEA, VS Code, Agile / Scrum, SDLC, Debugging, Troubleshooting, SOLID Principles, Design Patterns, Code Reviews

PROFESSIONAL EXPERIENCE

Software Engineer Intern | InternPro.ai Jan 2026 - Present

- Worked on modernizing legacy retail applications by migrating to **.NET 6**, simplifying long-term maintenance and improving cross-browser stability, increasing operational efficiency by 25%
- Designed and developed **ASP.NET Core** REST APIs by collaborating with stakeholders to gather requirements for high-volume inventory and order workflows
- Identified **SQL Server** performance bottlenecks and optimized queries using indexing strategies, reducing redundant database calls and improving retrieval efficiency
- Diagnosed recurring API failures and reduced them by ~25% through **DTO validation** and centralized **exception handling**, improving platform reliability and reducing disruptions in retail operations
- Implemented secure access using **JWT** authentication, OAuth2, and Role-Based Access Control within ASP.NET Core, strengthening service security and reducing authorization-related production issues
- Validated application workflows and business logic using **xUnit** and **integration testing**, reducing post-release defects and improving deployment stability

Associate Software Engineer | BNP Paribas Jun 2023 - Nov 2023

- Developed and optimized data retrieval and request handling logic for internal **trade settlements application** using caching strategies and indexing techniques, reducing response time by **20%** and infrastructure cost by **15%**
- Engineered **object-oriented microservice** with **Spring Boot** to forecast transaction volume and processing load using historical financial data; enabled efficient handling of over **10,000** transactions per minute
- Integrated the microservice with existing banking systems through **RESTful API** endpoints, connecting it to a **PostgreSQL** database for seamless communication and efficient data exchange
- Collaborated with production support to diagnose and resolve an average of **4 production issues per week**
- Conducted code reviews and contributed to SDLC processes in **Agile Scrum** using **Jira** and **Confluence**, improving performance and reducing release issues

Software Engineer Intern | HighRadius Jan 2022 - Apr 2022

- Built **React.js** based billing dashboards using **component-driven** architecture with **Tailwind CSS** and HTML, enabling efficient analysis of large-scale financial data
- Optimized frontend **performance** on high-volume datasets using lazy loading and memoization, reducing page load time by **~2sec**
- Integrated REST APIs and managed asynchronous data flow, enabling real-time updates and display of **10K+ financial records**
- Improved code quality and UI reliability using **Git** workflows, pull requests, and peer reviews, reducing defects across production releases

PROJECTS

Smart Email Assistant | Java, Spring Boot, Google Gemini, MySQL, Postman, AWS EC2

- Developed Spring Boot APIs integrating Gemini to generate context-aware email replies, improving response relevance through refined prompts and validation logic
- Built a Chrome extension integrating with Gmail, enabling real-time AI reply generation via a custom “AI Reply” button
- Designed end-to-end workflow by aligning with user requirements, enabling seamless email content processing and AI-generated response integration into the UI
- Implemented MySQL persistence for storing user interactions, used Postman for API testing and deployed backend services on AWS EC2

Recommendations Microservice | Python, Flask, Docker, Kubernetes, GitHub Actions, IBM Cloud

- Built a RESTful eCommerce recommendations microservice using Python and Flask, following REST API standards and Agile development practices
- Implemented CRUD operations, resource querying, and action endpoints while maintaining 97%+ unit test coverage using TDD and BDD frameworks
- Containerized the service with Docker and deployed it to IBM Cloud Kubernetes clusters across development and production environments
- Automated CI/CD workflows using GitHub Actions for unit tests, integration tests, Docker image builds, and multi-stage deployments

Chat Box using LangChain | RAG, LangChain, Python, LLMs

- Designed a Chat Box using LangChain and OpenAI with a 100+ page knowledge base, enabling context-aware responses across user queries
- Employed document splitting and built vector database with OpenAI embeddings for semantic search, improving retrieval accuracy and relevance
- Applied prompt engineering and a lightweight UI to generate answers to user queries, maintaining chat history for conversational continuity and context retention

Multi-Agent Financial Analysis System | Python, LangGraph, OpenAI GPT-4.1, MCP, Bright Data

- Architected a multi-agent system with 4 specialized agents for market data retrieval, sentiment analysis, and decision synthesis, coordinated via a supervisor
- Integrated real-time financial data using MCP-based orchestration, enabling agents to dynamically access market signals and news without custom pipelines
- Built an LLM-driven pipeline combining external data sources to generate structured financial insights, reducing manual analysis effort

EDUCATION

Arizona State University Tempe, Arizona
Master of Science in Information Technology, GPA: 4.0/4.0 Jan 2024 - Dec 2025

SRM Institute of Science & Technology Chennai, India
Bachelor of Technology in Computer Science and Engineering, GPA: 3.6/4.0 Jun 2019 - May 2023

CERTIFICATIONS

- AWS Certified Developer - Associate
- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner